

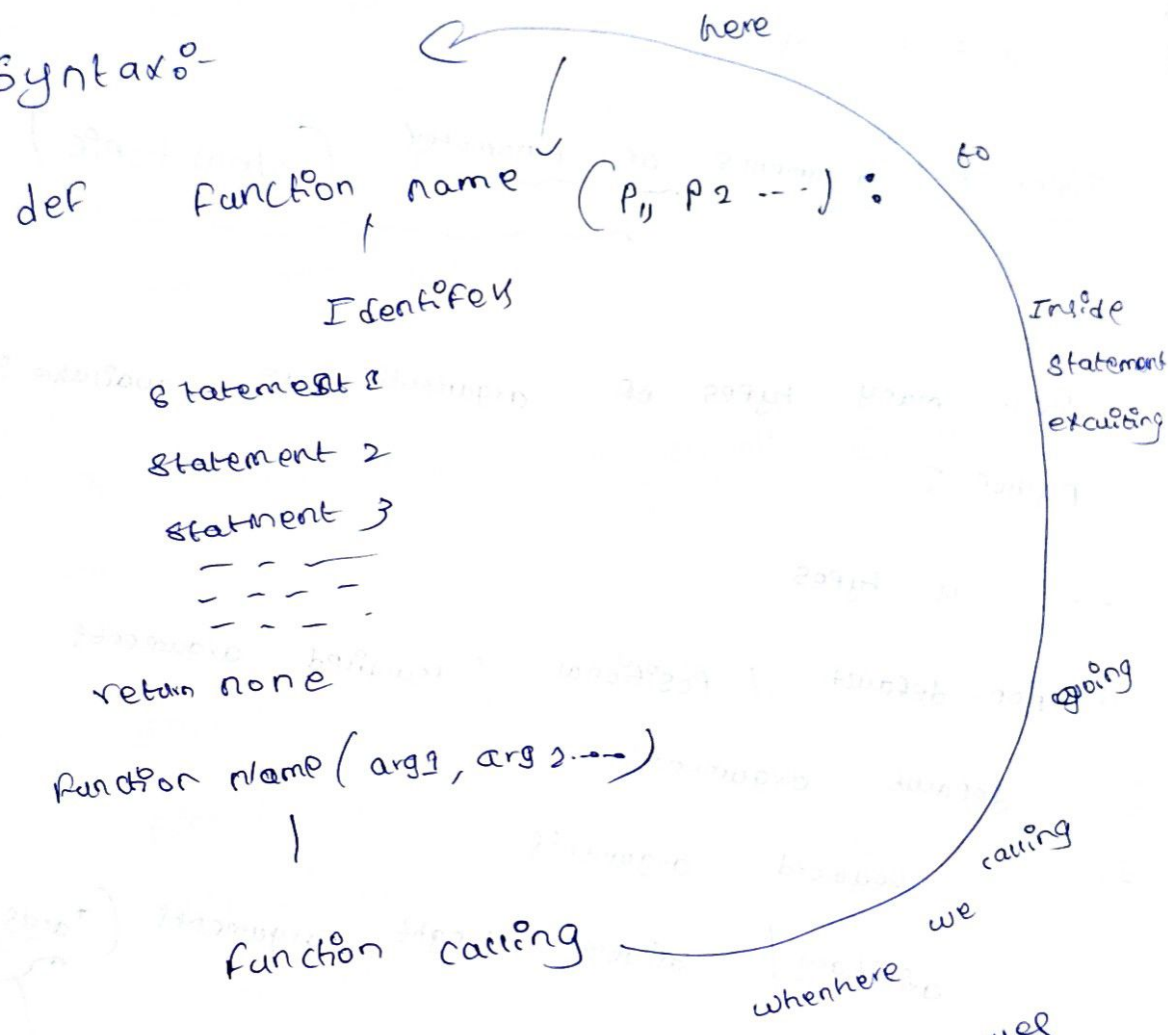
Functions + + Impor

- * A function is a block of organized reusable code that is used to perform a single or related task
- * A function is a block of code that performs a specific and well defined task
- * It organizes the code into a logical way to perform a certain task
- * Function is a syntax of structure in which we represents the reusable business logic -
- by using "def" keyword we can create function in python
- * every function is called by a certain name and block of code inside the function gets executed when we call the function
- * If you have a block of code that gets involved more than once put it into a function

• This is because function allows us to use a block of code repeatedly in different sections of a program

• Functions help to break down long and complex programs into smaller and manageable segments and enhance program readability and comprehensibility

Syntax:-



def function_name (optional - parameter)
 ...
 return none

function name (optional parameter)
 variables or variables

How many types of functions are available in Python

1. Predefined --> print(), type(), id(), int(), float(), input(), len() ---

2. userdefined --> f1(), f2(), message(), display(), add(), etc.

Types of arguments or parameters (New topic)

How many types of arguments are available in Python?

--- 4 types

1) non-default / positional / required arguments

2) default arguments

3) keyword arguments

4) arbitrary / variable length arguments (+args, **kwargs)

Popular interview question

1) non-default / positional / required arguments

```
def display (a,b)

    print ('value of a : ', a)
    print ('value of b : ', b)

display (10,20)
```

Default arguments

```
def display (a,b) (a=10, b=20);

    print ('value of a : ', a)
    print ('value of b : ', b)

display (10,20) ()
```

Working arguments { with both non-default & default

```
def display (a, b=20):

    print ('value of a : ', a)
    print ('value of b : ', b)

display (15, 25)
a=15      b=25
```

③ Key word arguments

```
def employee (eid, ename, salary, dept-name):

    print ('Employee id : ', eid)
    print ('Employee name : ', ename)
    print ('Employee salary : ', salary)
    print ('Employee id : ', eid)
```